Energy 34 (2009) 1855–1863

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Factors affecting electricity demand in Athens, Greece and London, UK: A comparative assessment

B.E. Psiloglou [a,*], C. Giannakopoulos [a], S. Majithia [b], M. Petrakis [a]

[a] Institute of Environmental Research & Sustainable Development, National Observatory of Athens, I. Metaxa & Vas. Pavlou, GR 15236, P. Pentelh, Athens, Greece
[b] National Grid Control Center, St. Catherine's Lodge, Berkshire RG41 5BN, England, UK

ARTICLE INFO

Article history:
Received 28 January 2009
Received in revised form
20 July 2009
Accepted 27 July 2009
Available online 20 August 2009

Keywords:
Electricity demand
Thermal comfort
Athens
London
Heating and cooling demands

ABSTRACT

This paper presents a comparison of the characteristics of the electricity demand for London, UK and Athens, Greece and explores its relationship with both climate and non-climate related factors. Year-to-year trends for both cities are identified, associated mainly with economic, social and demographic factors. In addition, several other effects such as weekly and holiday effects, unrelated to weather conditions are detected and are examined in comparison for the two cities. Gross National Product (GNP) per capita has been employed to aid de-trend the data in order to isolate the weather/climate influence on electricity demand. Temperature has been found to play the most important role in controlling the electricity load demand, especially for Athens. For both cities electricity demand peaks in winter but for Athens a second significant peak is apparent in the summer, not present in London. Thermal comfort levels for both cities have also been identified and compared. The optimum ambient temperature for low levels of electricity demand is found to be 20 °C for Athens and 16 °C for London. Using the approach of cooling and heating degree-days, it was identified that the sensitivity of the system electricity-demand/air-temperature is greater during the cold period of the year for both cities.

© 2009 Elsevier Ltd. All rights reserved.

1. Introduction

Several economic activities are affected by weather fluctuations. One of the most sensitive is the electricity market, since power demand is closely linked to air temperature. The sensitivity of energy demand to weather stems from the fact that that produced electricity must be instantly consumed.

Various studies have investigated the influence of ambient air temperature, most times represented by heating and cooling degree-days, on electrical energy consumption [1–10]. However, temperature is not the only variable considered in the literature. Other primitive independent variables, such as relative humidity, clearness index, cloudiness, rainfall, solar radiation and wind speed [11–15], and derived variables including latent enthalpy-days, temperature–humidity index, Steadman's indoor apparent temperature, cooling radiation-days and clothing insulation units '*clo*' [8,16–18] have also been used by other researchers for the development of statistical models for energy consumption. In many cases modeling of electric energy consumption is multivariate, consisting in a mix between climate and other important economic factors. The main constituents of these economic factors are energy prices, income, Gross National Product (GNP), import and export values and energy demand index [17,19–25]. Population and production in total manufacturing, together with temperature, have also been used by Bessec and Fouquau [26] for modeling monthly electricity consumption to a panel of 15 member states of the European Union over the last two decades.

Lai et al. [27] tried to model the electricity consumption in the Asian gaming and tourism center of Macao SAR. As Macao is a very small city and tourists spend plenty of their time in an indoor environment, such as casinos, restaurants and hotel rooms, they proposed a model equation of monthly electricity consumption incorporating (except monthly mean air temperature and population) two more commercial-related determinants, the number of tourists per month and the hotel room occupancy. Also, in order to avoid abnormal data appearing in February (28 or 29 days) and other months (30 or 31 days) they proposed the number of days per month as a third determinant.

Co-integration techniques for establishing long- and short-term energy demand relationships with climate factors have also been performed [12,25]. Moral-Carcedo and Vicéns-Otero [28] in their analysis of the effects of temperature on the electricity demand in Spain, proposed a logistic smooth threshold regression model with the temperature as a threshold variable. This allows the relationship

* Corresponding author. Tel.: +30 210 8109133; Fax: +30 210 8103236.
E-mail address: bill@meteo.noa.gr (B.E. Psiloglou).

0360-5442/$ – see front matter © 2009 Elsevier Ltd. All rights reserved.
doi:10.1016/j.energy.2009.07.033

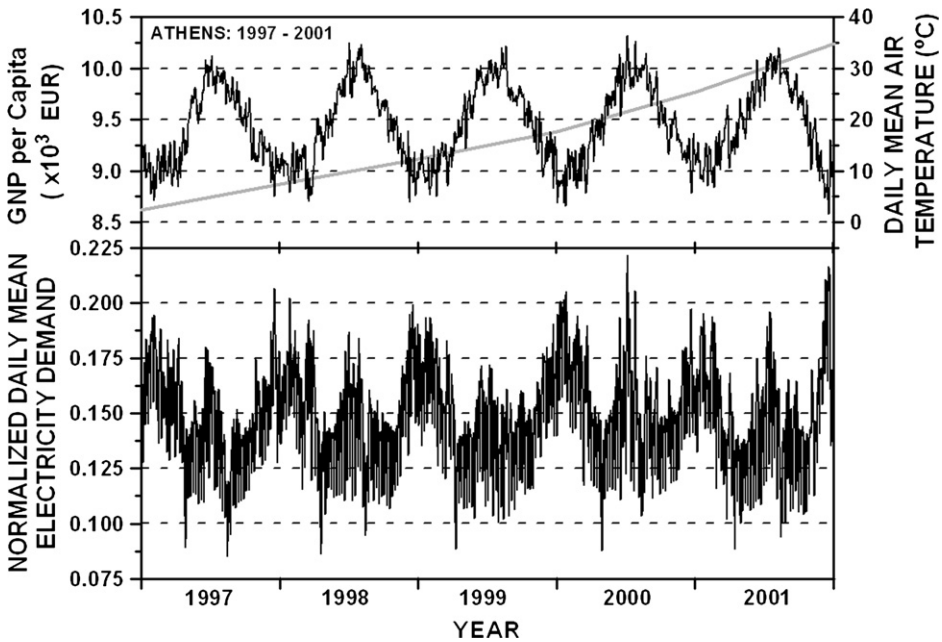

Fig. 1. Mean daily air temperature and electricity demand, normalized using the GNP per capita (gray line), for Athens, Greece, for the period 1997–2001.

between electricity consumption and temperature to depend on the level of the threshold variable i.e. the temperature. The same technique was used in the analysis by Bessec and Fouquau [26].

Daily electricity demand in countries throughout the world shows a clear seasonal pattern. Average daily electricity demand in most EU-15 countries, historically shows a single peak during winter months [29]. Only Spain, Portugal, Italy and Greece show an additional peak during summer months [5,29,30]. Demand patterns with only summer peaks may be found outside of Europe, e.g. as observed in Bangkok [31] and Hong Kong [1]. The seasonal pattern results from the fluctuating influx of solar radiation and the varying economic activity throughout the year in the Northern hemisphere [3].

Time series analysis of daily electricity demand data reviles a "U" shape relation between outdoor temperature and electricity demand [3,5,10,11,15,25,26,28,32,33]. This non-linearity refers to the fact that both increases and decreases of temperature, linked to exceeding certain "threshold" temperatures, increase the demand for electricity. According to Henley and Peirson [2], this response is caused by the differences between the ambient or outdoor temperature and the comfort or indoor temperature. When the differential between outdoor and indoor temperatures increases, the starting-up of the corresponding heating or cooling equipment immediately raises the demand for electricity. Naturally, the curve of the response of demand to temperatures depends especially on the climate characteristics of the geographical area to which the

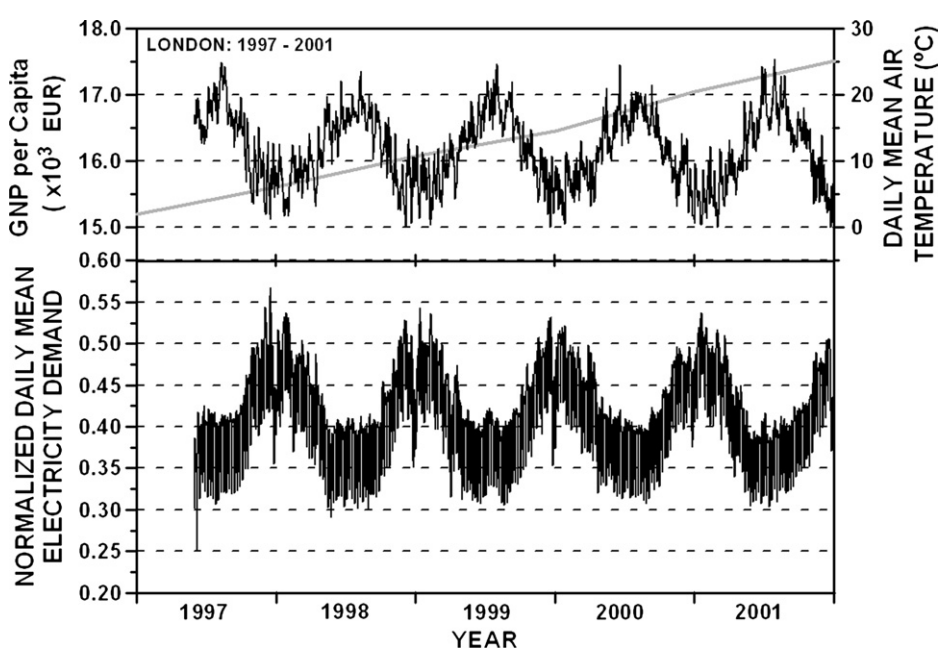

Fig. 2. Mean daily air temperature and electricity demand, normalized using the GNP per capita (gray line), for London, UK, for the period 1997–2001.

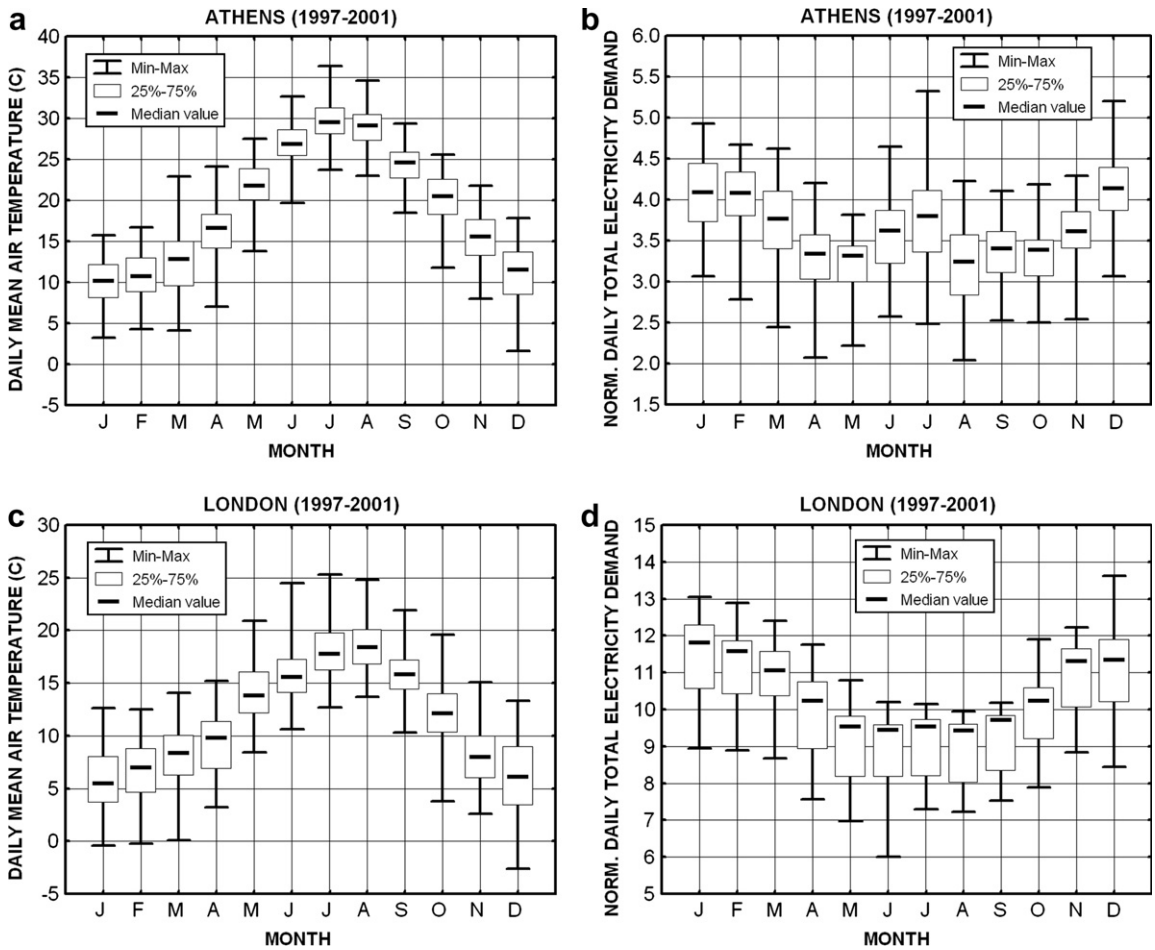

Fig. 3. Box-whisker plots for daily mean air temperature and normalized daily total electricity demand, for (a, b) Athens, Greece and (c, d) London, UK, for the period 1997–2001.

demand data refer, which condition the type of equipment is installed, and the conditions under which it will start to operate. Thus, in especially hot areas, there will be hardly any response to cold, and conversely in relatively cold or temperate regions, there will be little or no response to heat. Also, the notion of indoor temperature is not a parameter that can be considered stable or immutable, since variables such as income, consumer preferences or energy prices, can cause variations in it.

This paper aims to provide a comparative analysis of the relationship between electricity load demand and air temperature for Athens, Greece and London, UK. We initially identify the general trends that govern the electricity demand in the areas of Athens and London (yearly, seasonal, daily, hourly). We then seek to determine the relationship between electricity demand and ambient monthly or daily air temperature. In dealing with the non-linearity of the temperature effect, the most frequent posture in the literature is to segment temperature variations in terms of the functions CDD and HDD. The cooling and heating degree-days concepts (CDD and HDD) have been employed by truncation of the temperature series at constant base temperature levels which were found to be appropriate for the weather characterization in Athens and London. Up till now, mostly empirical studies of the relation between electrical energy consumption and climate have been performed, which also tended to be published only in internal reports of utility companies. Our study aims to bridge this gap in literature and complement the few existing studies of Valor et al. [5] for Spain, Giannakopoulos and Psiloglou [30] for Athens and the studies of Sailor and Muñoz [11] and Sailor [4] for the USA, attempting for the first time a comparison of the electricity demand characteristics between two European cities: London in the Northwest of Europe and Athens in the Southeast using high resolution real time energy data. It should be noted that it is the first time that such a comparative study using hourly energy data is attempted since in the literature [26] typically monthly data are used. Moreover, in this study, we use a methodology with indexes [5,30] that permits to compare different levels of consumption patterns for the two cities of Athens, Greece and London, UK.

2. Data description

Hourly electrical energy consumption data were made available by the *Public Power Corporation of Greece* for the greater Athens area, and by the *National Grid of UK*, for the greater London area. These data refer to total *hourly* residential and commercial electricity consumption only, in kWh, spanning the period from 1997 through 2001. This period was chosen as it was the only time period that energy consumption data were available for both cities. Additionally, meteorological data from the station at the National Observatory of Athens (NOA) for Athens, and from the United Kingdom Meteorological Office for London were also used for the needs of this study. The meteorological data from both stations used in this study are hourly values of temperature, covering the same time period as the electricity demand data.

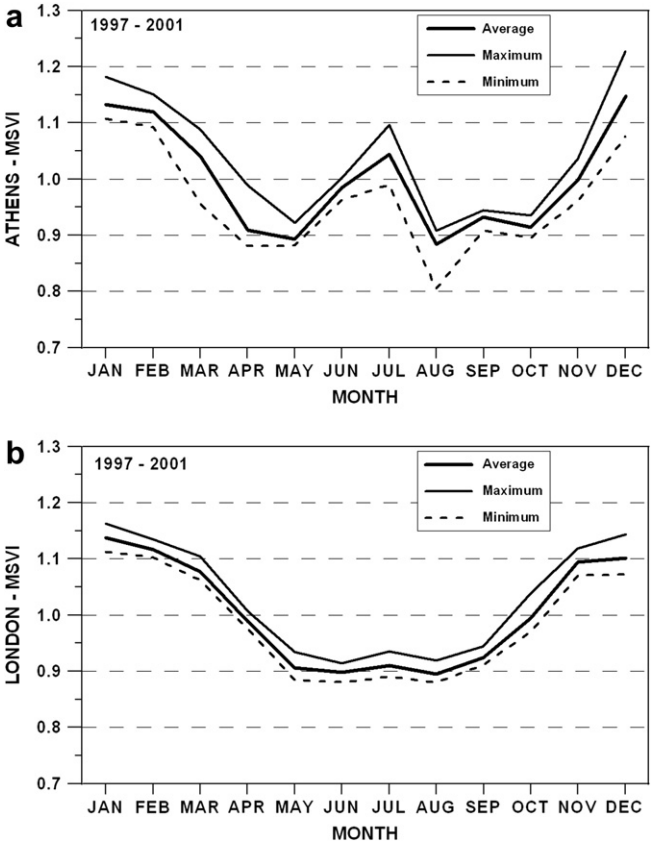

Fig. 4. Variation of MSVI index according to month of the year for electricity demand, for (a) Athens, Greece and (b) London, UK, for the period 1997–2001.

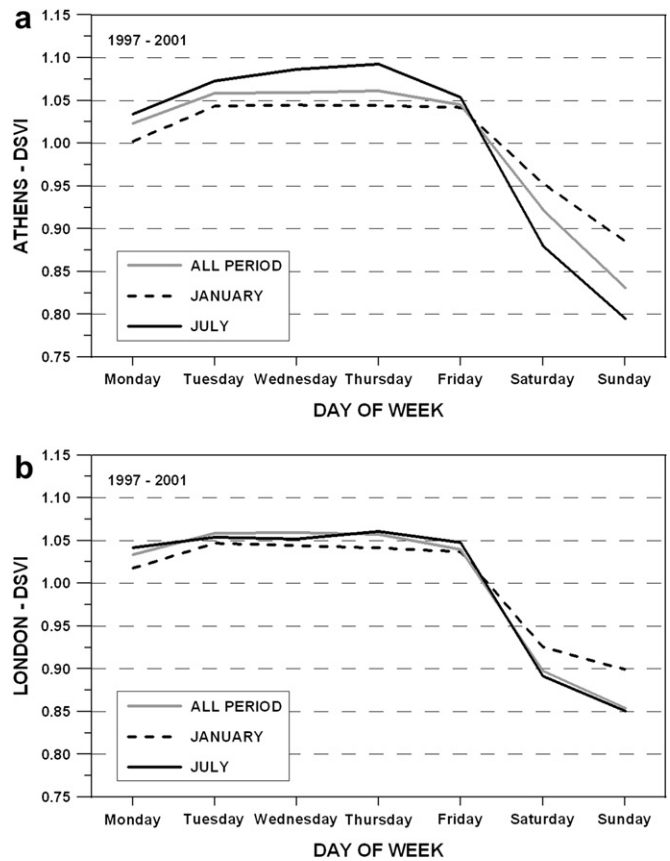

Fig. 5. Variation of DSVI index according to day of the week for electricity demand, for (a) Athens, Greece and (b) London, UK, for the period 1997–2001.

For both cities, a slight upward trend in electricity demand is present throughout the examined years (not shown here) mainly associated with economic growth (see [30] for Athens). For this reason, Gross National Product (GNP) per capita has been employed in both cases to aid de-trend the data in order to isolate the weather/climate influence on electricity load demand.

Mean daily variation of air temperature, gross national product (GNP) for the greater Athens and London areas and normalized electricity demands for the period 1997–2001 are depicted in Figs. 1 and 2. A seasonal trend is apparent in both cities, influenced by the prevailing weather conditions with peaks in electricity demand coinciding with extremes in temperature. The link between hot weather and increased electricity demand is also evident in other cities worldwide, apart from Athens [1,17,31].

Fig. 3 shows the statistics using box-whisker plots for daily mean air temperature and normalized daily total electricity demand, respectively, for both sites and for the whole period examined.

For the Athens case, the largest deviations from the mean in electricity demand occur during the summer and winter months when hot and cold temperature extremes respectively are more common. The same is true for London, but only during winter months.

3. Electricity load demand and air temperature

3.1. Daily, weekly and monthly electricity load variability

In this section we compare the temporal cycles (monthly, diurnal, hourly) in electricity load demand for both cities. The use of seasonal variation indexes has been chosen to appropriately study these cycles.

The monthly seasonal variation index (MSVI) can be defined as [5,30]:

$$\mathrm{MSVI}_{ij} = E_{ij}/\overline{E_j} \tag{1}$$

where MSVI_{ij}, the index value for month i in year j; E_{ij}, the monthly electricity load demand for month i in year j; $\overline{E_j}$ the monthly average electricity load demand for year j (the mean of the 12 values of E_{ij} for year j).

Fig. 4 depicts the monthly seasonal profile of MSVI for electricity demand from 1997 through 2001, for Athens, Greece and London, UK. The average value of MSVI yields the relative behaviour throughout the months of the year whereas the maximum and the minimum values of MSVI reveal the deviations from the mean behaviour. For the greater Athens area, the behaviour of MSVI shows an increased demand of electricity in January, followed by a gradual decrease until May, which is associated with the decrease in demand for heating. As we move into summer, there is, as expected, an increasing tendency, mainly due to extensive air-conditioning use. The tendency continues throughout the summer until September with the exception of August, when a significant fall in demand is evident, due to the fact that the major part of the population of the greater Athens area is on summer vacation during this month. Electricity demand also slightly falls in October, which is representative of a transient season from summer to winter and gradually increases again in the coming months to reach a maximum in December. The electricity load values in December are even higher than those in January and February (which are

colder months than December) due to the increased electricity needs during the Christmas festive period.

For the greater London area, there seem to be two periods in the year, a cold period where electricity demand levels rise as temperatures fall and a relatively warmer period, with no apparent need for air-conditioning, where electricity levels are kept relatively constant at lower levels. Contrary to Athens, summer peak electricity demand levels associated with high temperatures do not exist for London as temperatures do not reach the values of Athens. Furthermore, there are no transitional seasons for London. The lowest values of electricity demand for London are in June, July and August while before and after these months electricity levels start to rise to reach their highest values in December, January and February.

To explore the daily fluctuation of electricity load demand, the daily variation index (DSVI) is defined as [5,30]:

$$\mathrm{DSVI}_{ijk} = E_{ijk}/\overline{E_{jk}} \tag{2}$$

where DSVI_{ijk}, the value of the index for day i of week j of year k; E_{ijk}, is the electricity load demand for the particular day; $\overline{E_{jk}}$, the mean daily electricity load demand for week j of year k.

Fig. 5 depicts the variation of DSVI according to the day of the week. For both cities, it is clear that electricity demand is significantly lower during weekends (especially on Sundays) due to the reduced economic activities this period. Lower levels of consumption are also present on Mondays compared to the other days of the week because of the inertia caused by the reduced economic activities during the weekends. Moreover, electricity demand also decreases on the holidays that fall between Monday and Friday and also on the days that follow a holiday or are placed between two holidays (again due to the inertia).

Fig. 5 also demonstrates the seasonal characteristics of DSVI. More specifically, it indicates that electricity demand is higher during weekdays in July whereas in January there are higher values of electricity demand during weekends. This reflects the fact that during weekends in July, people are away from the offices and tend to go outdoors or out of the city boundaries. The opposite holds true for January.

Given that our data include hourly values of electricity demand, the hourly seasonal variation index (HSVI) was also calculated for both cities, which shows the diurnal variation of electricity demand. From the hourly values, the average diurnal variation for each month was calculated (24 average hourly values each month). In this way, an average 24-h period is available for each month, so for each year 12 such 24-h periods are present. HSVI is therefore defined as [30]:

$$\mathrm{HSVI}_{ijk} = E'_{ijk}/\overline{E'_{jk}} \tag{3}$$

where HSVI_{ijk}, the value of the index for hour i of the average 24-h period, of month j in year k; E'_{ijk}, the electricity load demand for a certain hour; $\overline{E'_{jk}}$, the mean monthly electricity load demand for month j in year k.

In Fig. 6, the hourly variation during a 24-h period is evident (lines with gray color) for both cities. For Athens, the first maximum close to midday is due to the extensive use of electricity both for household (heating, cooking) and business (office heating, server and PC usage) needs during the working hours of the day. The second maximum is due to the use of lighting and heating/cooling using additional heaters or air-conditioners during the late afternoon and early evening hours.

In London electricity demand is kept constantly high during office hours, i.e. from around 9 a.m. to 7 p.m., with two less pronounced maxima, one around midday and the second in the early evening hours. The first maximum is due to home and office needs as in Athens, while the second is probably due to the fact that around 5–7 p.m. some people still stay at their offices while others go out for recreation (cafes, restaurants, bars). Therefore, the second maximum is due both to office and recreational activities needs.

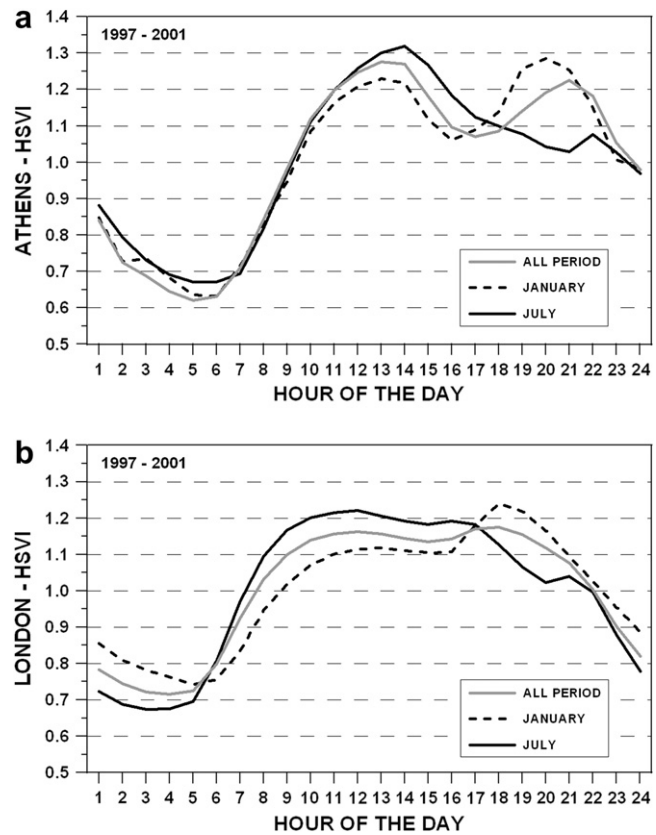

Fig. 6. Diurnal variation of HSVI index for electricity demand, for (a) Athens, Greece and (b) London, UK, for the period 1997–2001.

It is worth noting the differences in HSVI when studying it in different seasons, most notably summer and winter. As a characteristic example we also present in Fig. 6, the months of January and July.

For Athens during January, there are two distinct maxima in electricity demand, one close to midday (due to heating, cooking or office needs) and another in the evening. The evening maximum reflects needs for lighting (because of shorter daylight period), extra heating and home entertainment (e.g. television) as most people stay indoors. The pattern in July is different. Apart from the midday maximum, which occurs approximately at the same hours as the January one, the evening maximum occurs much later and is much less pronounced than the January one. Moreover, in the late afternoon/early evening hours, there is a decrease in demand which reflects the fact that most people like to stay outdoors for a dinner or other recreational activities and return home later.

For London, the hourly pattern of electricity demand is somewhat different from that in Athens. During January, there is only one distinct peak in the evening hours. In July, the evening peak disappears, but demand appears to be high throughout the daylight hours and especially around midday.

3.2. Relationship between electricity load and air temperature

The variation of normalized daily electricity demand (using GNP data as described in Section 2) and mean daily air

temperature for the whole of 2001, for Athens, Greece and London, UK, is plotted as an example in Fig. 7. Electricity demand exhibits not only a persistent trend and seasonality but also, given the daily frequency of the data analysed, the predominance of the working day effect is patent. Electricity demand presents a clear weekly variation with minimum values on Sundays (the deep troughs apparent in the electricity data time series every 7 days) and maximum values close to midweek. This pattern is occasionally interrupted on isolated holidays falling in the middle of the week. Regular variations in electricity demand, discussed in Section 3.1, are also apparent in the electricity demand plot (black line), in both cases.

In contrast, the mean daily air temperature presents a much less regular variability, and follows a typical seasonal trend. This variability is closely linked to electricity demand, whose maximum values correlate with the extreme values of air temperature (maximum or minimum). During December, January and February, the maximum values of electricity demand are related to the appearance of the lowest temperatures. For Athens, during the transient season of March–April, when air temperatures are constantly rising, electricity demand levels are kept nearly constant until about May.

Contrary to Athens, summer peak electricity demand levels associated with high temperatures do not exist for London as temperatures do not reach the values of Athens. Also, there are no transitional seasons for London.

Fig. 8 shows that electricity demand is greatly reduced during weekends and holidays compared to the demand during working days. The smallest values of electricity demand exist during the long Easter weekend (Good Friday to Easter Monday) as well as on other fixed holidays such as Christmas for both cities or the 15th of August for Athens irrespective of the daily mean air temperature.

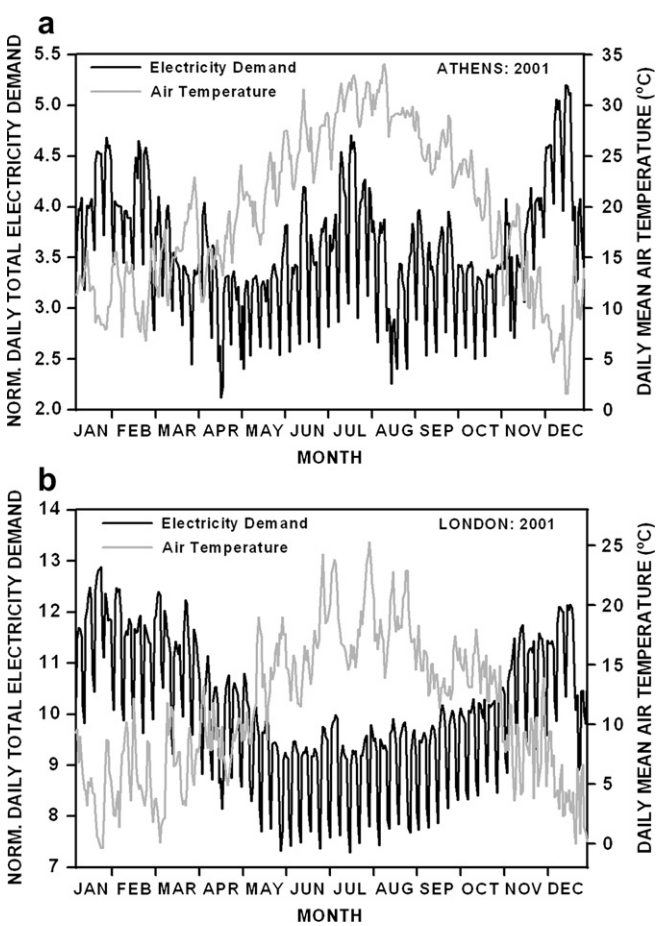

Fig. 7. Variation of normalized daily total electricity demand and mean daily air temperature, for (a) Athens, Greece and (b) London, UK, for the year 2001.

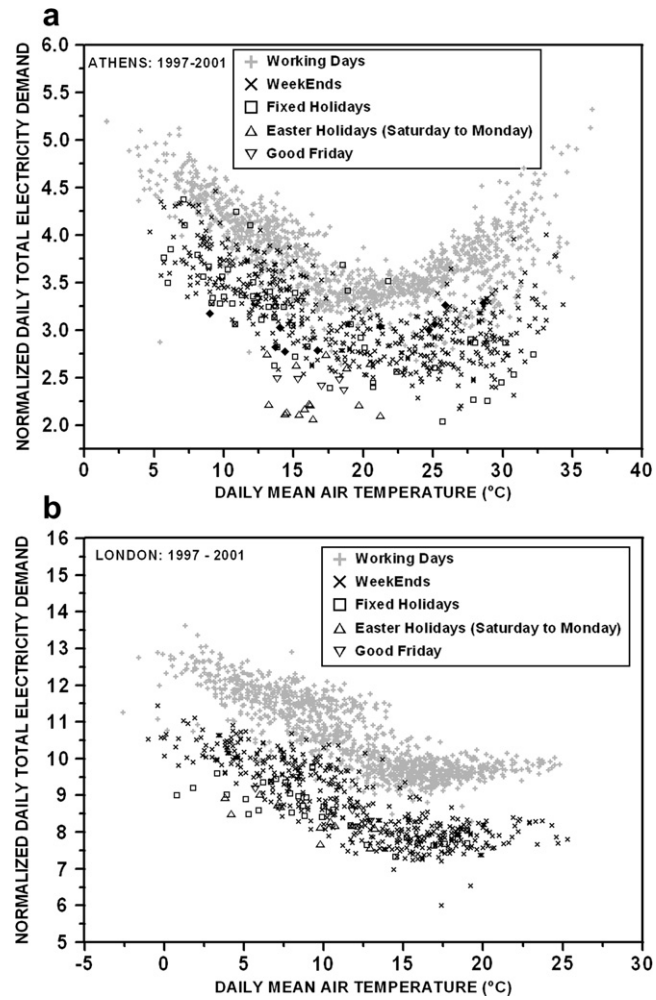

Fig. 8. Scatter plot of normalized daily total electricity demand versus daily mean air temperature, for (a) Athens, Greece and (b) London, UK, for the period 1997–2001.

Since weekends/holidays seem to correlate less closely with air temperature, holidays were separated from working days and their correlation with air temperature was explored separately (Fig. 9).

Fig. 9 shows that, for the Athens case, the relationship between electricity demand and air temperature is not linear and presents one minimum and two maxima. The minimum value appears around 20 °C and it is the temperature that can be used for the calculation of heating and cooling degree-days (see Section 4). Around this temperature, there exists an area where electricity demand shows no sensitivity to air temperature. Outside this area, electricity demand increases with the increase (due to air-conditioning needs) or decrease (due to extra heating needs) of air temperature. However, there are certain limits beyond which electricity demand does not increase any further. This happens in both branches of the curve corresponding to winter and summer. This may be due to the limiting power of conditioning systems and to the insolating capacity of the buildings [5]. Also, the summer branch is less developed and remains lower than the winter branch.

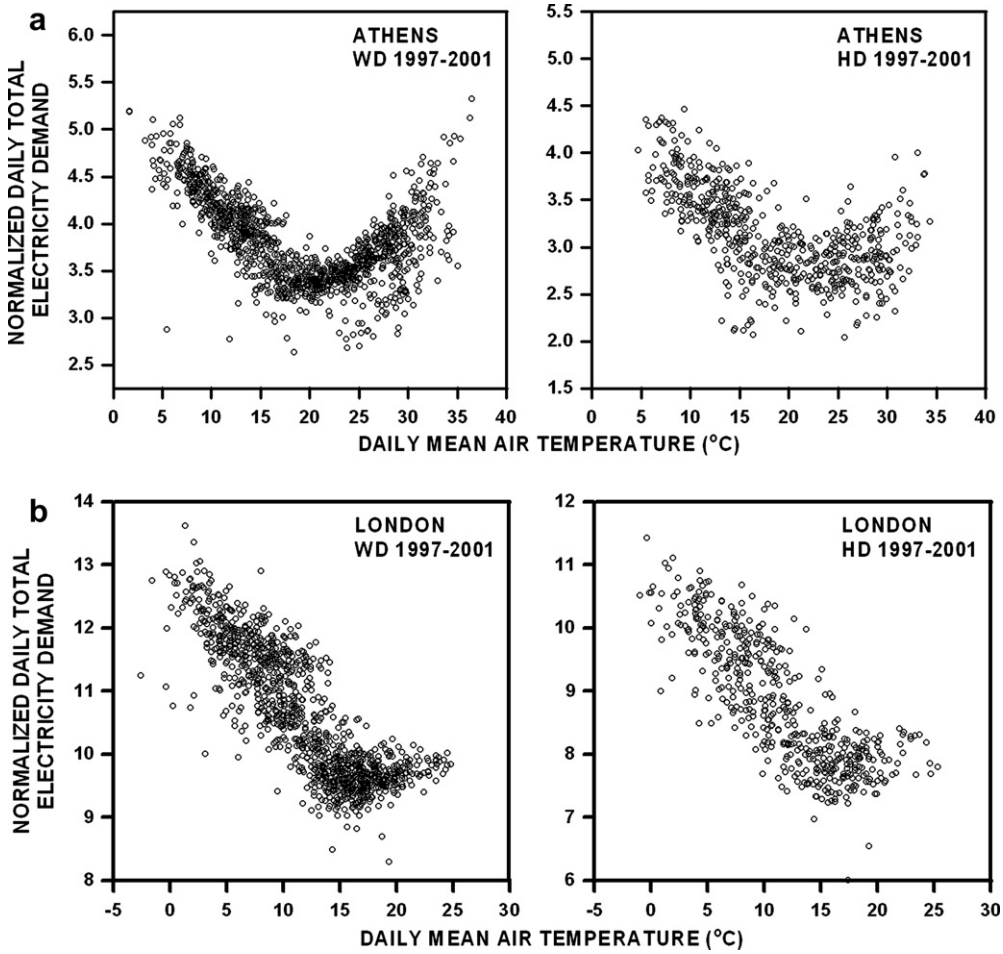

Fig. 9. Scatter plots of normalized daily total electricity demand versus daily mean air temperature, for (a) Athens, Greece and (b) London, UK, for the period 1997–2001 on working days (WD) and weekends/holidays (HD).

London, UK, also exhibits a non-linear relationship between air temperature and electricity demand but with only one minimum and one well defined maximum. The minimum values of electricity demand appear to be around 16 °C and this is the temperature that can be used for the calculation of heating and cooling degree-days for London. Above this temperature, electricity levels for London tend to rise slightly but below this temperature electricity levels increase significantly.

4. Heating and cooling demands

Since the electrical energy - temperature relationship is non-linear and has two branches, it would be more convenient for further data interpretation to separate these two branches. The easiest way to achieve this is to use the idea of degree-days (DD) (firstly introduced by Thom [34–36]), which is defined as the difference of mean daily temperature from a base temperature. As base temperature (T), we define the ambient air temperature at which solar and internal (from electric lighting, equipment and people) gains offset heat losses through building insulation [37] and at this point electricity demand shows no sensitivity to air temperature.

The base temperature is related to the desired indoor temperature. Thus, in turn, is influenced by factors such as lifestyle and income, and hence varies in both space and time.

In order to have positive values of DD index in both summer and winter branches, the definition of two indices is used: heating degree-days (HDD) and cooling degree-days (CDD).

For the calculation of the HDD and CDD indices, the following equations were used:

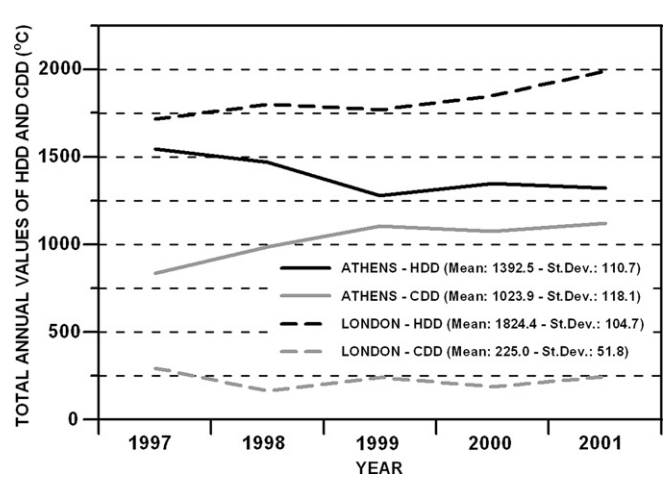

Fig. 10. Total annual variation in heating degree-days (HDD) and cooling degree-days (CDD) for the period 1997–2001, using 20 °C for Athens, Greece, and 16 °C for London, UK, as base temperatures.

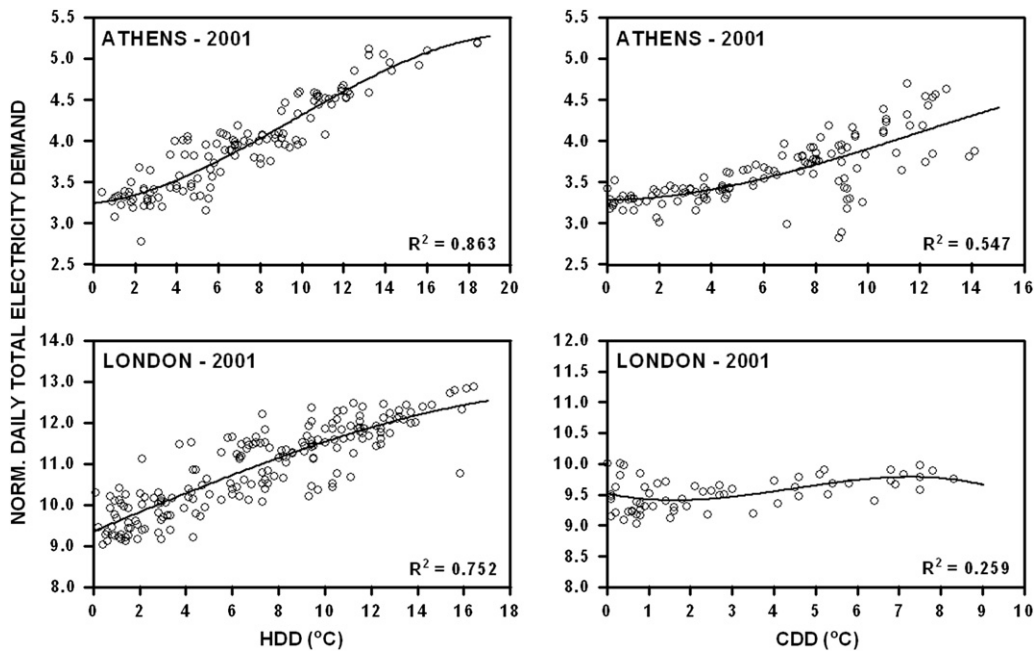

Fig. 11. Normalized daily total electricity demand with HDD and CDD for working days only of the year 2001, for Athens, Greece, and London, UK, using 20 °C and 16 °C as base temperatures, respectively. The black line represents the third order polynomial fit to the data and R^2 is the corresponding correlation coefficient.

$$\mathrm{HDD} = \max\left(T^* - T, 0\right) \tag{4}$$

$$\mathrm{CDD} = \max\left(T - T^{**}, 0\right) \tag{5}$$

where T is the mean daily temperature as calculated from the hourly available meteorological data, and T^* and T^{**} are the base temperatures for HDD and CDD, respectively, which can be either the same or different, allowing for a temperature interval within electricity demand is irresponsive to temperature variations. These two indices reflect in combination the number of days on which the temperature stands below or above the thresholds of cold and heat, and by how many degrees.

HDD (CDD) is a measure of the severity and duration of cold winter (hot summer) conditions in terms of the outdoor dry-bulb air temperature, an indication of the sensible heating (cooling) requirements for the particular location. The unique base temperature of 18 °C is commonly used to calculate HDD and CDD, especially in the analysis of the impact of weather on electricity demand [3,5,37–42].

In our case, the base temperature can be identified using the correlation plots of electricity demand with air temperature in Fig. 9 for the working days. Base temperature should be the temperature where electricity demand is at its minimum. This temperature is approximately equal to 20 °C for the case of Athens and 16 °C for London for the entire period with available data (1997–2001). Therefore, $T^* = T^{**} = 20$ °C or 16 °C for Athens, Greece, and London, UK, respectively.

Fig. 10 presents the total annual variation of the HDD and CDD in order to identify possible changes in the variation of the two indices for the period under examination (1997–2001), for Athens, Greece and London, UK. Values in Fig. 10 show that total values of HDD are larger than those of CDD. This would indicate that we have more cold than warm days per year. However, CDD and HDD values are sensitive to the selection of the base temperature. If a lower base temperature had been chosen for Athens, Greece (such as 18 °C, as also used in the literature) warm days would be much more and cold days fewer. The opposite is true for London, UK, if a greater base temperature had been chosen.

For the Athens case, another feature evident from Fig. 10 is a declining trend for HDD and an increasing trend for CDD. This trend of degree-days indices could be due to a warming in the Athens area experienced in the last decade [43]. For London, UK, an increasing trend in HDDs and a stable pattern in CDDs reflecting the temperature variations in the period chosen (1997–2001, Fig. 2).

Fig. 11 shows the daily electricity demand in relation to the values of CDD and HDD as they were calculated using 20 °C as base temperature for Athens, Greece, and 16 °C for London, UK, for 2001. The two branches, summer and winter ones, have been separated in two independent correlation functions.

From Fig. 11 it becomes apparent that correlation is higher for the heating degree-days, signifying that the response of the system is faster during the cold period of the year.

5. Conclusions

This work presents a comparative study of electricity demand and air temperature for two cities: Athens, Greece and London, UK, for the period 1997–2001. For both cities electricity demand peaks in winter but for Athens a second significant peak is apparent in the summer, which is not found in London. The winter peak is associated with low temperatures in both cities and the summer peak is linked to hot temperatures and increased air-conditioning use in Athens. Both cities exhibit lower electricity demand levels during public holidays and weekends (especially Sundays and summer weekends). The hourly variation of electricity demand levels has two maxima in Athens around midday and around evening but is constantly high throughout office hours for London.

In both cities, the relationship between electricity demand and air temperature is not linear and presents one minimum and two maxima for Athens but only one minimum and one well defined maximum for London. The minimum value of electricity load demand for Athens appears around 20 °C whereas around 16 °C for London. Around these temperatures in both cities, there exists an

area where electricity demand shows no sensitivity to air temperature. Outside this area, electricity demand levels increase with the increase of air temperature sharply for Athens but only slightly for London. Electricity demand levels also increase with the decrease of air temperature substantially for both cities. For both cities, it was found that the response of the system electricity load demand/air temperature is faster during the cold period of the year.

The knowledge of the energy characteristics of the two cities has important policy implications for the power industry. Indeed, as the economic structure of each city is different and unique, power planning and management should take key climatic, demographic and economic factors into consideration to address the pressing issue of energy saving. This will aid each local city government to establish policies to encourage users by replacing old electrical equipment for energy saving. If feasible, introducing renewable energy systems should also be encouraged in the light of increased levels of energy demand due to climate change. Climate change may, in turn, result in summer electricity demand peaks, seen already in southern European cities (such as Athens) to occur even in northern European cities, affecting in this way, generation capacity, maintenance scheduling and pricing.

Acknowledgements

This study was performed in the framework of EU projects ENSEMBLES (www.ensembles-eu.org) under contract GOCE-505539 and CIRCE (www.circeproject.eu) under contract GOCE-036961, the support of which are greatly acknowledged.

The authors are grateful to the Strategy and Planning Department of the Public Power Corporation of Greece for providing the electricity demand data for the greater Athens area.

References

[1] Al-Zayer J, Al-Ibrahim A. Modelling the impact of temperature on electricity consumption in the eastern province of Saudi Arabia. J Forecasting 1996;15:97–106.
[2] Henley A, Peirson J. Non-linearities in electricity demand and temperature: parametric versus non-parametric methods. Oxford Bull Econ Stat 1997;59:149–62.
[3] Pardo A, Meneu V, Valor E. Temperature and seasonality influences on Spanish electricity load. Energy Econ 2002;24:55–70.
[4] Sailor DJ. Relating residential and commercial sector electricity loads to climate – evaluating state level sensitivities and vulnerabilities. Energy 2001;26:645–57.
[5] Valor E, Meneu V, Caselles V. Daily air temperature and electricity load in Spain. J Appl Meteor 2001;40:1413–21.
[6] Matzarakis A, Balafoutis C. Heating degree-days over Greece as an index of energy consumption. Int J Climatol 2004;24:1817–28.
[7] Amato AD, Ruth M, Kirshen P, Horwitz J. Regional energy demand response to climate change: methodology and application to the commonwealth of Massachusetts. Climate Change 2005;71:175–201.
[8] Thatcher MJ. Modeling changes to electricity demand load duration curves as a consequence of predicted climate change for Australia. Energy 2007;32:1647–59.
[9] Xiao N, Zarnikau J, Damien P. Testing functional forms in energy modeling: an application of the Bayesian approach to the US electricity demand. Energy Econ 2007;29:158–66.
[10] Cancelo JR, Espasa A, Grafe R. Forecasting the electricity load from one day to one week ahead for the Spanish system operator. Int J Forecast 2008;24:588–602.
[11] Sailor DJ, Muñoz JR. Sensitivity of electricity and natural gas consumption to climate in the U.S.A. Methodology and results for eight states. Energy 1997;22:987–98.
[12] Bard EA, Nasr GE. On the relationship between electrical energy consumption and climate factors in Lebanon: co-integration and error-correction models. Int J Energy Res 2001;25:1033–42.
[13] Lam JC, Wan KKW, Cheung KL, Yang L. Principal component analysis of electricity use in office buildings. Energy Build 2008;40:828–36.
[14] Lam JC, Tang HL, Li DHW. Seasonal variations in residential and commercial sector electricity consumption in Hong Kong. Energy 2008;33:513–23.
[15] Ihara T, Genchi Y, Sato T, Yamaguchi K, Endo Y. City-block-scale sensitivity of electricity consumption to air temperature and air humidity in business districts of Tokyo, Japan. Energy 2008;33:1634–45.
[16] Yan YY. Climate and residential electricity consumption in Hong Kong. Energy 1998;23:17–20.
[17] Lam JC. Climatic and economic influences on residential electricity consumption. Energy Convers Mgmt 1998;39:623–9.
[18] Howden S, Crimp S. Effect of climate and climate change on electricity demand. MODSIM 2001 international congress on modelling and simulation; 2000. p. 655–60.
[19] Arsenault E, Bernard JT, Car CW, Genest-Laplante E. A total energy demand model of Quebec. Energy Econ 1995;17:163–71.
[20] Zarniko JA. Re-examination of the causal relationship between energy consumption and gross national product. J Energy Develop 1997;21:229–39.
[21] Beenstock M, Goldin E, Nabot D. The demand for electricity in Israel. Energy Econ 1999;21:168–83.
[22] Nasr GE, Bard EA, Dibeh GL. Econometric modeling of electrical energy consumption in postwar Lebanon. Energy Econ 2000;22:627–40.
[23] Mohamed Z, Bodger P. Forecasting electricity consumption in New Zealand using economic and demographic variables. Energy 2005;30:1833–43.
[24] Ozturk HK, Ceylan H. Forecasting total and industrial sector electricity demand based on genetic algorithm approach: Turkey case study. Int J Energy Res 2005;29:829–40.
[25] Mirasgedis S, Sarafidis Y, Georgopoulou E, Kotroni V, Lagouvardos K, Lalas DP. Modeling framework for estimating impacts of climate change on electricity demand at regional level: case of Greece. Energy Convers Manage 2007;48:1737–50.
[26] Bessec M, Fouquau J. The non-linear link between electricity consumption and temperature in Europe: a threshold panel approach. Energy Econ 2008;30:2705–21.
[27] Lai TM, To WM, Lo WC, Choy YS. Modeling of electricity consumption in the Asian gaming and tourism center-Macao SAR, People's Republic of China. Energy 2008;33:679–88.
[28] Moral-Carcedo J, Vicéns-Otero J. Modeling the non-linear response of Spanish electricity demand to temperature variations. Energy Econ 2005;27:477–94.
[29] Hekkenberg M, Benders RMJ, Moll HC, Schoot Uiterkamp AJM. Indications for a changing electricity demand pattern: the temperature dependence of electricity demand in the Netherlands. Energy Policy 2009;37:1542–51.
[30] Giannakopoulos C, Psiloglou BE. Trends in energy load demand for Athens, Greece: weather and non-weather related factors. Clim Res 2006;31:97–108.
[31] Wangpattarapong K, Maneewan S, Ketjoy N, Rakwichian W. The impacts of climatic and economic factors on residential electricity consumption of Bangkok Metropolis. Energy Build 2008;40:1419–25.
[32] Ruth M, Lin AC. Regional energy demand and adaptations to climate change: methodology and application to the state of Maryland, USA. Energy Policy 2006;34:2820–33.
[33] Franco G, Sanstad A. Climate change and electricity demand in California. Climatic Change 2008;87:139–51.
[34] Thom HSC. Seasonal degree days statistics for the United States. Mon Weather Rev 1952;80:143–9.
[35] Thom HSC. The rational relationship between heating degree days and temperature. Mon Weather Rev 1954;82:1–6.
[36] Thom HSC. Normal degree days above any base by the universal truncation coefficient. Mon. Weather Rev 1962;94:461–5.
[37] Eto JH. On using degree-days to account for the effects of weather on annual energy use in office buildings. Energy Build 1988;12:113–27.
[38] Badescu V, Zamfir E. Degree days, degree hours and ambient temperature bin data from monthly average temperatures (Romania). Energy Convers Mgmt 1999;40:885–900.
[39] Hart M, de Dear R. Weather sensitivity in household appliance energy end-use. Energy Build 2004;36:161–74.
[40] Le Comte DM, Warren HE. Modelling the impact of summer temperatures on national electricity consumption. J Appl Meteor 1981;20:1415–9.
[41] Quayle RG, Diaz HF. Heating degree-days data applied to residential heating energy consumption. J Appl Meteor 1980;19:241–6.
[42] Isaac M, van Vuuren DP. Modeling global residential sector energy demand for heating and air conditioning in the contex of climate change. Energy Policy 2009;37:507–21.
[43] Founda D, Papadopoulos KH, Petrakis M, Giannakopoulos C, Good P. Analysis of mean maximum and minimum temperatures in Athens from 1897 to 2001 with emphasis to the last decade: trends, warm and cold events. Global Planet Change 2004;44:27–38.